17. Using ROS 2 Rviz2

1. Introduction to Rviz2

During robot development, if we analyze the various functions only at the data level, it is difficult to quickly understand the effect of the data. Take the robot model as an example: we need to know what the model we designed looks like, and where the many coordinate frames inside the model are located during motion.

Another example is robotic arm motion planning and autonomous navigation of mobile robots. We want to see the environment around the robot, the planned path, and of course sensor information such as cameras, 3D cameras, and LiDAR. Data is used for computation, while visualization is what people actually look at.

Therefore, data visualization can greatly improve development efficiency. Rviz2 is exactly such a data visualization software for robot development: robot models, sensor information, environment information, and more can all be handled here.

2. Preparation

First, load the ROS 2 Jazzy environment in the terminal where the simulation will be started:

```
source /opt/ros/jazzy/setup.bash
```

- If you have a physical robot, you can start rviz on the robot's main control terminal to practice the content of this lesson. If you do not have a physical robot, you can launch a TurtleBot3 simulated robot through Gazebo simulation to simulate topics such as LiDAR and camera, which facilitates the following data visualization.

Note: The following installation steps are optional. If you have a physical robot, you can directly use the radar information of the real robot after setting up multi-machine communication; you can choose to use the real robot's radar or a virtual simulated robot. The following content is intended for users without a physical robot.

- This lesson takes a simulated robot as an example to teach the visualization functions of rviz2. Whether it is a real robot or a simulated robot, the operation flow of rviz2 is the same.
- Install the TurtleBot3 simulator package

```
sudo apt install ros-${ROS_DISTRO}-turtlebot3*
```

- Install the ROS and Gazebo bridge tool

```
sudo apt install ros-${ROS_DISTRO}-ros-gz
```

- Set the TurtleBot3 robot type environment variable

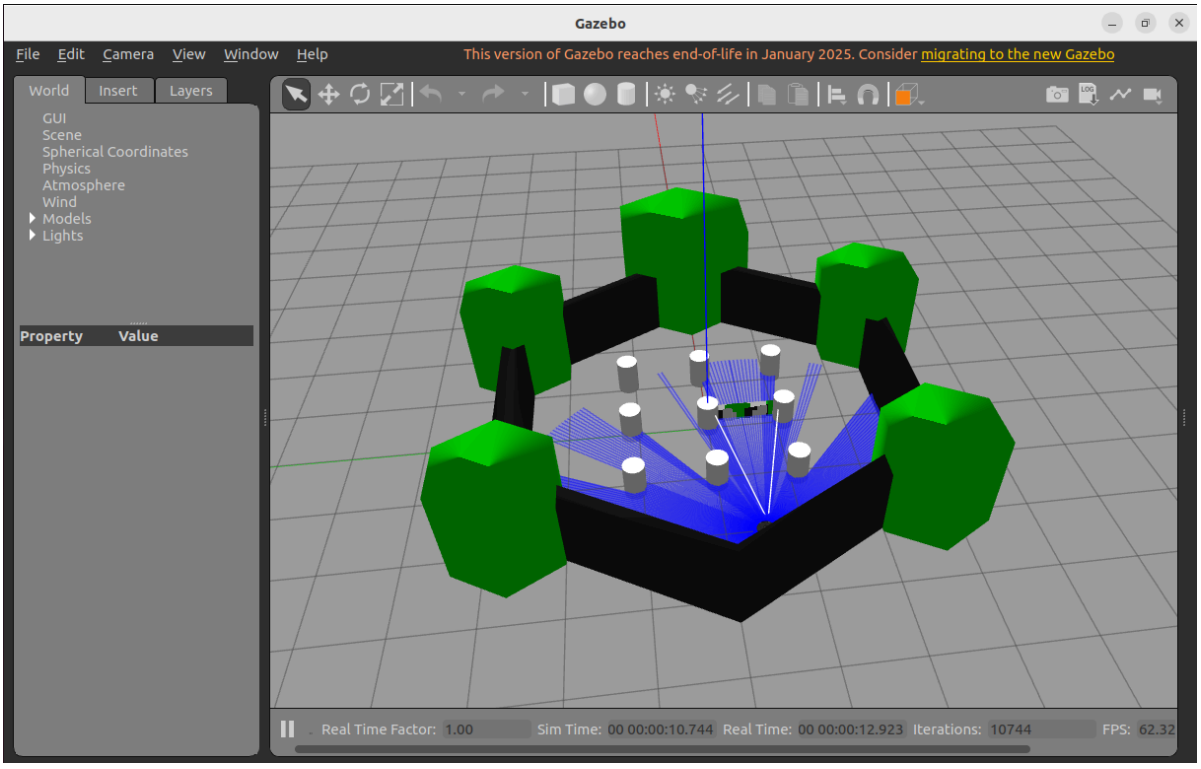
```
export TURTLEBOT3_MODEL=waffle
```

`TURTLEBOT3_MODEL` must be valid in the same terminal where the simulation is started. You can first run `echo $TURTLEBOT3_MODEL` to confirm that it returns `waffle`; otherwise the launch file will report `'TURTLEBOT3_MODEL'`.

- Start the Gazebo simulation environment

```
ros2 launch turtlebot3_gazebo turtlebot3_world.launch.py
```

It is recommended to run it in the VNC graphical desktop. SSH or RDP terminals without a correct `DISPLAY` may fail to show the Gazebo GUI; devices with small memory should also close other Gazebo processes first to avoid EGL errors or insufficient memory. Do not start TurtleBot3 and other Gazebo worlds in this tutorial at the same time.

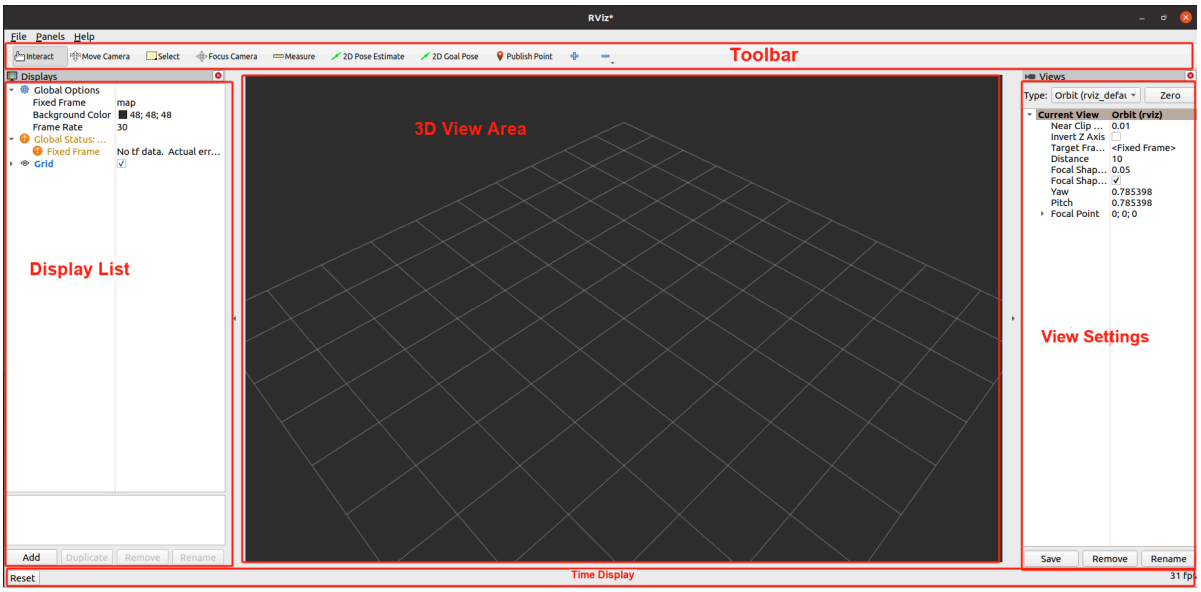


3. Starting Rviz2

Open a terminal and use the following command to start it:

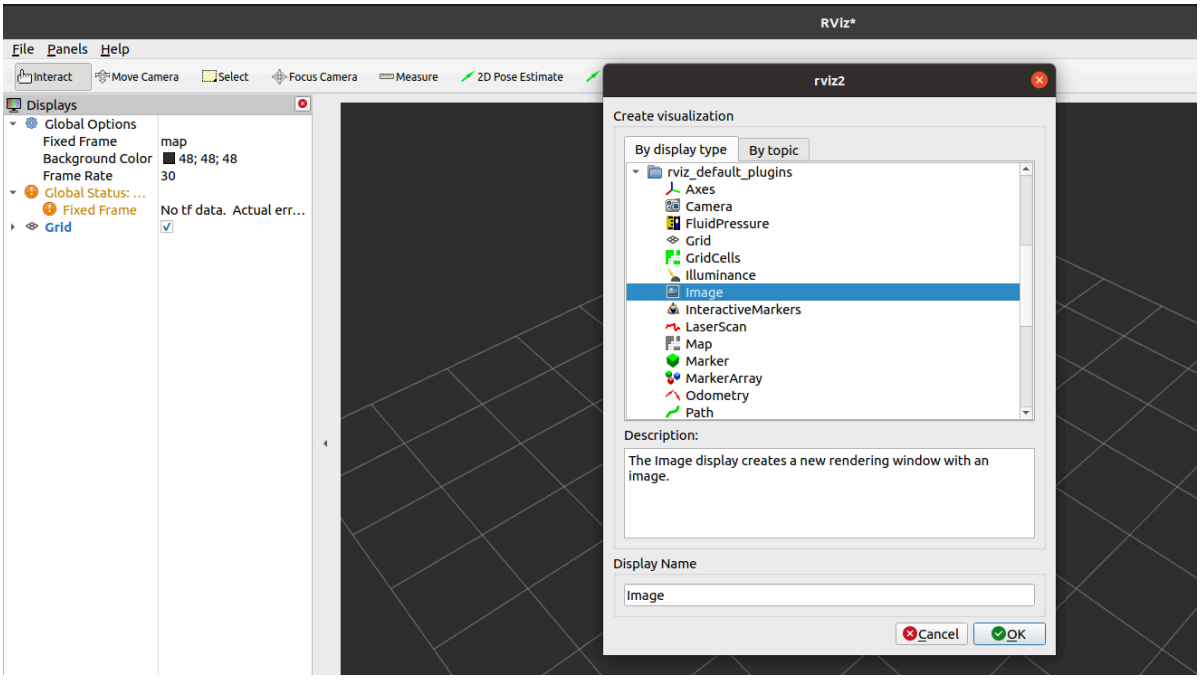
```
rviz2
```

If starting in Docker, make sure the GUI display is enabled.

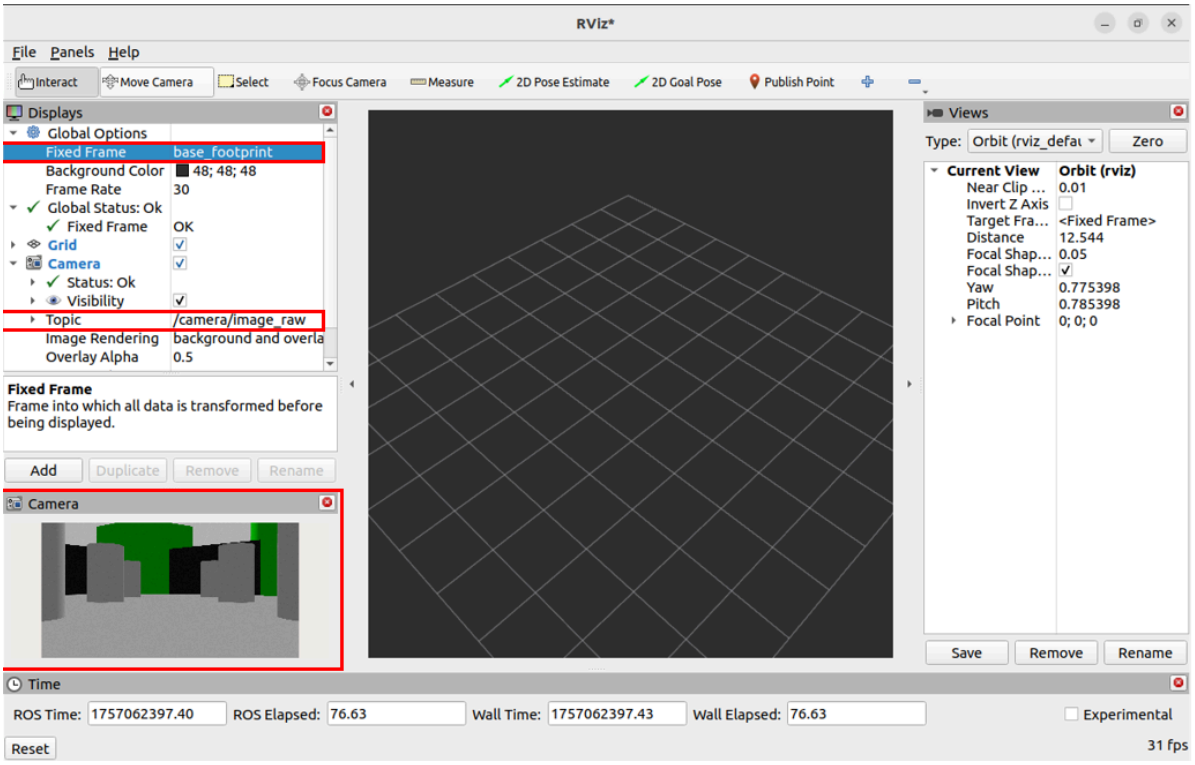


4. Image Data Visualization

In the Displays window on the left, click "Add", find the Image display item, and confirm with OK to add it to the display list. Then configure the image topic subscribed by this display item, and you can see the robot's camera image.

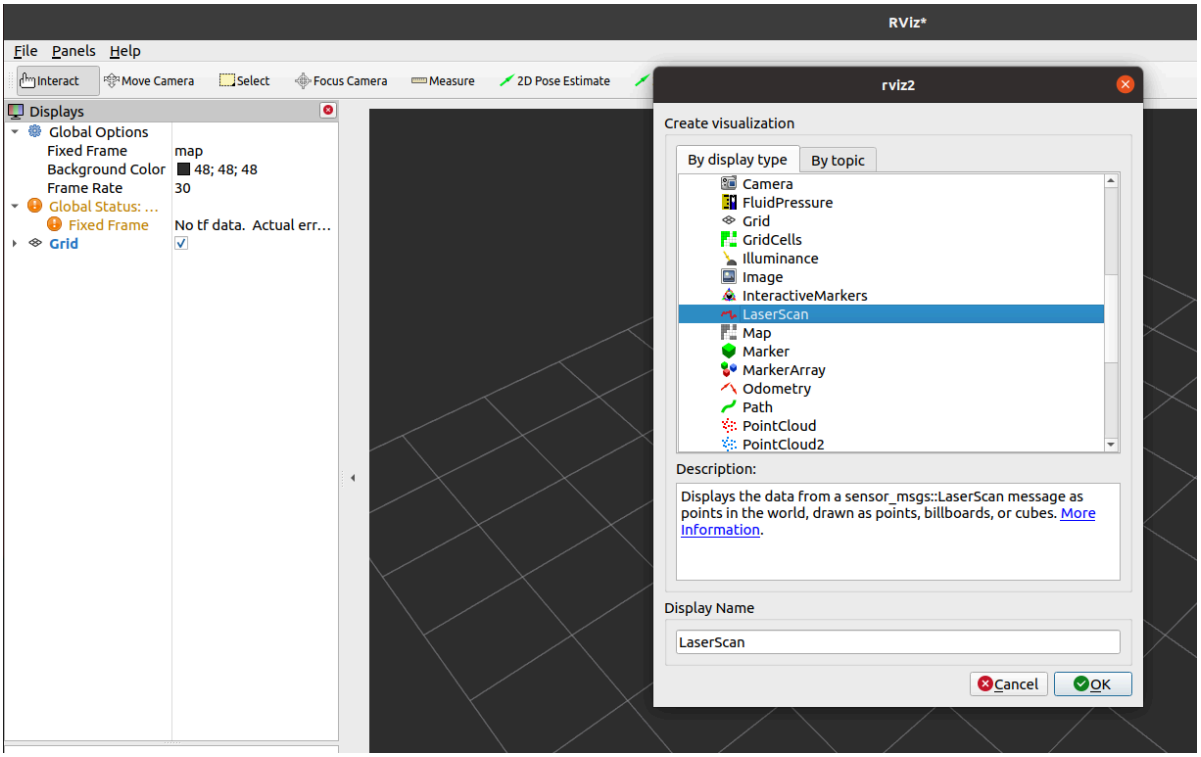


- Select `Fixed Frame` as the `base_footprint` coordinate frame to prevent coordinate transform errors
- Select the camera color image topic `/camera/image_raw`
- Now you can see the current view of the simulated robot in the Camera window

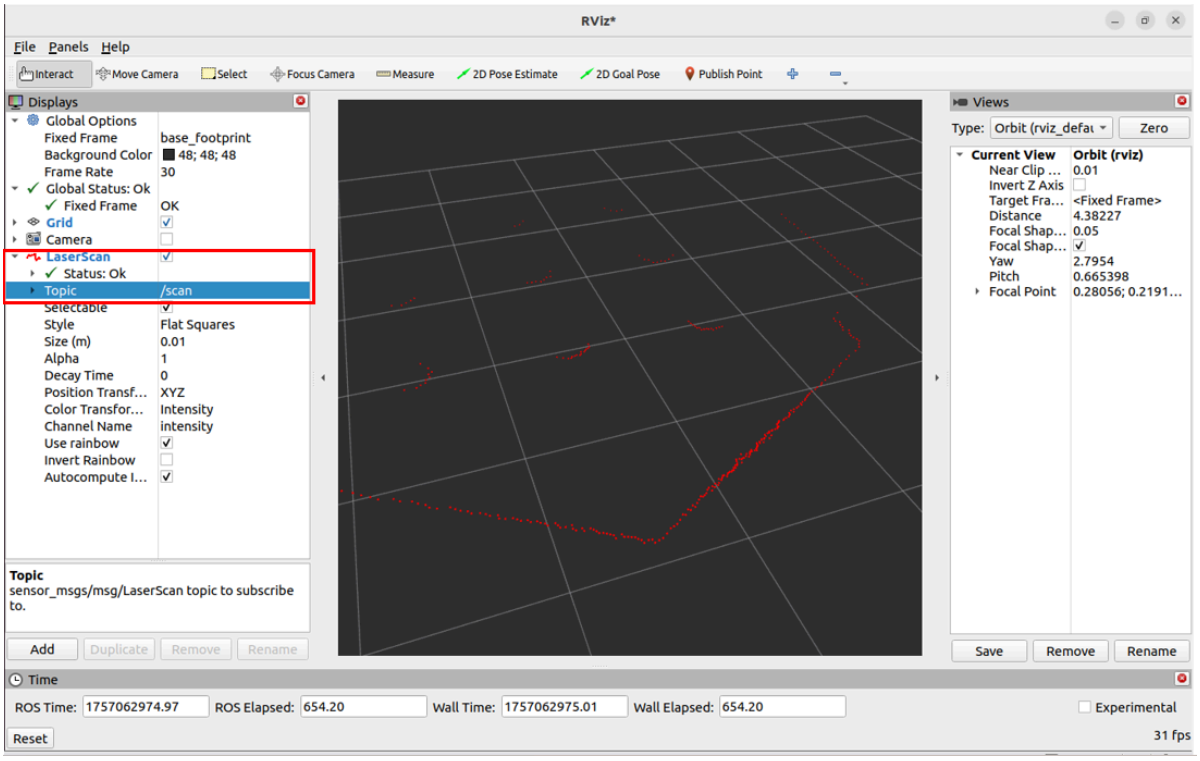


5. LiDAR Data Visualization

In the Displays window on the left, click "Add", select Laserscan, and then configure the subscribed topic name. The laser points are then visible.

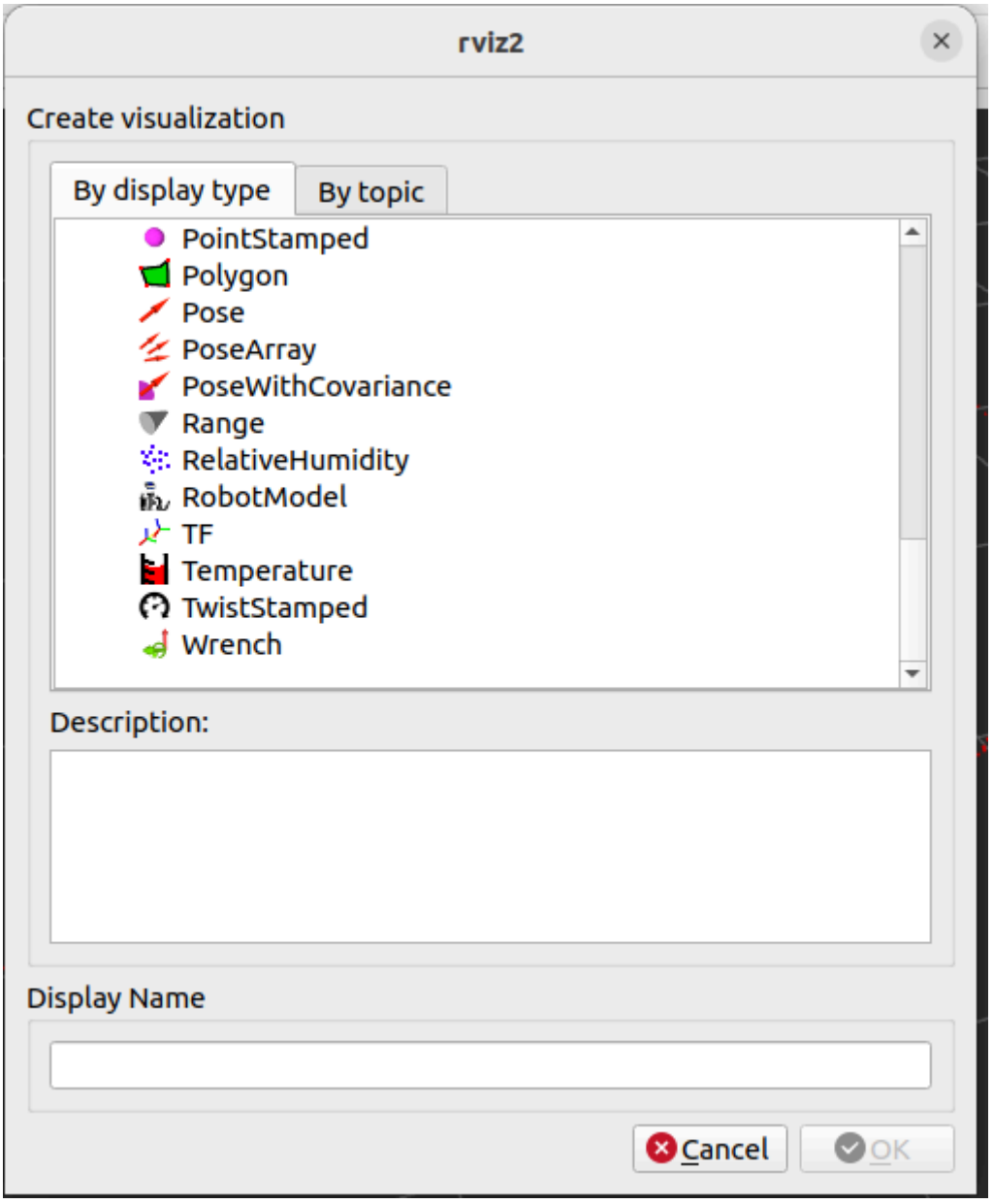


- For the `Topic` of `LaserScan`, select `/scan`
- Now you can see the point cloud outline of the LiDAR

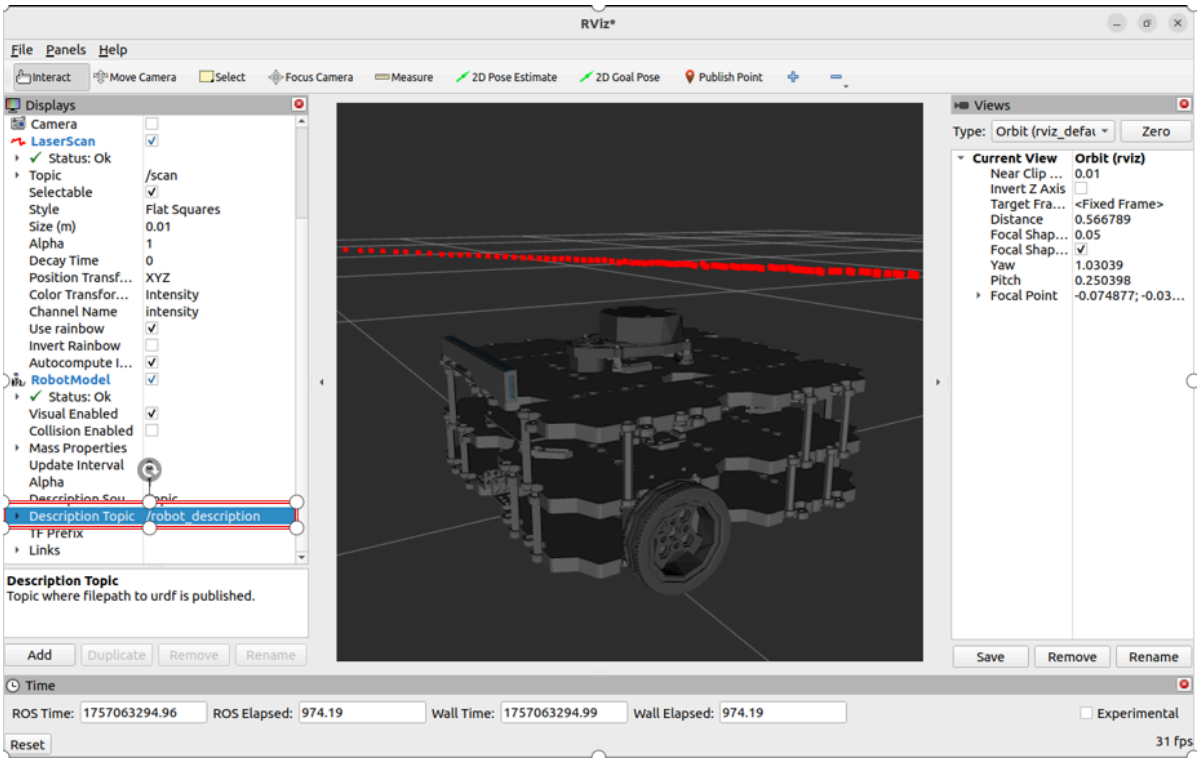


6. Robot Model Visualization

In the Displays window on the left, click "Add", and select RobotModel

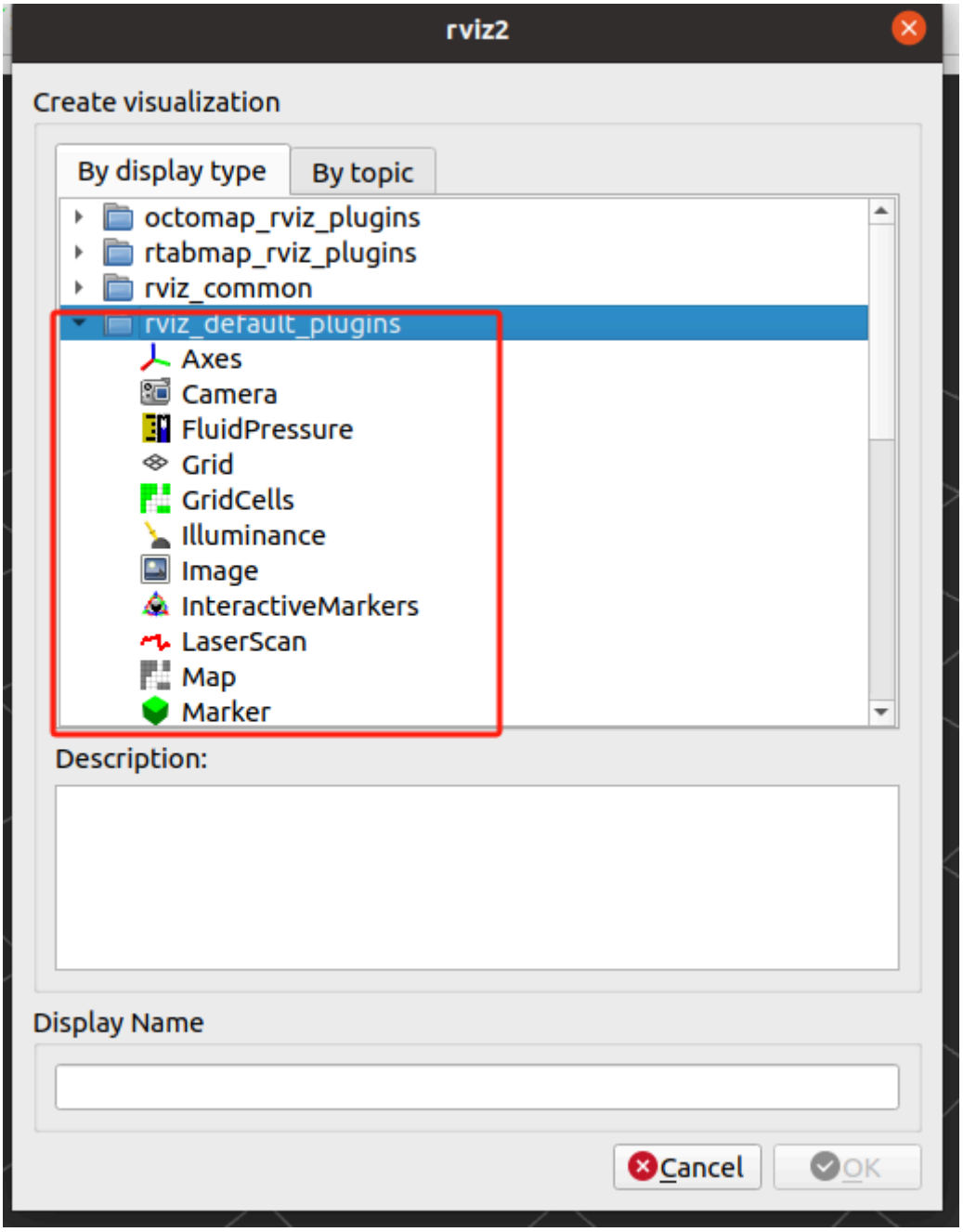


- In the robot's `DescriptionTopic` option, select the topic `/robot_description`
- Now you can see the visualized robot model in Rviz2



7. Other Data Visualizations

rviz_default_plugins lists many commonly used data visualization plugins; you can try them one by one.


rviz2
Create visualization
By display type
By topic
octomap_rviz_plugins
rtabmap_rviz_plugins
rviz_common
rviz_default_plugins
Axes
Camera
FluidPressure
Grid
GridCells
Illuminance
Image
InteractiveMarkers
LaserScan
Map
Marker
Description:
Display Name
Cancel
OK